Finite σ-Equivariant Row-Graded Magmas on AG$(2,3)$: Parameter Space, Intrinsic Selection, and Linearization

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Abstract

We study the finite parameter space of σ-equivariant row-graded magmas on $M = S \times S$, $S = \mathbb{F}_3$, equivalently the point set of AG$(2,3)$, with left-row output and a fixed same-row off-diagonal Steiner rule. The parameter space has normal form

$$\Omega' \cong S^{18} \times S^3 \cong S^{21}, \qquad |\Omega'| = 3^{21} = 10\,460\,353\,203.$$

Two finite enumeration theorems first map the ambient space. The associativity spectrum satisfies

$$63 \leq \text{Assoc} \leq 597,$$

with exact endpoint loci and partition function. The path-resolved Hamming defect satisfies

$$\min H_{tot} = -2268, \qquad \max H_{tot} = 7302, \qquad \max_{N_-=0} H_{tot} = 7020.$$

Inside this parameter space, four internal finite conditions

$$\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad (r,r)^2 = (r,r), \qquad \text{StabMaps}_R = \{C, J\}$$

select a unique point, the right-row projection magma RP. After selection, RP has a finite linearization profile: rank-three left translations, a 51-element left-translation semigroup, an endomorphism and intertwiner classification, the binary part of the term clone, of size 630, a row-sum contraction matrix realization, a Hamming-defect operator identity, associated compact Lie groups, and a signed symplectic lift of the C/J right-stabilizer geometry.

The theorem package is organized as

$$\text{finite parameter space} \Longrightarrow \text{intrinsic finite selection} \Longrightarrow \text{linearization of } \mathsf{RP}.$$

Contents

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Introduction and main theorem

Object

Let

$$S = \mathbb{F}_3, \qquad M = S \times S.$$

The first coordinate is the row and the second coordinate is the column. The paper studies binary operations on M satisfying left-row output, the fixed same-row off-diagonal Steiner rule, and equivariance under the simultaneous shift $\sigma(r,c) = (r+1, c+1)$. The normal coordinates are

$$(A, B, d) \in S^9 \times S^9 \times S^3,$$

where A and B encode the two ordered cross-row representatives and d encodes the same-point diagonal. The resulting finite parameter space is

$$\Omega' \cong S^{18} \times S^3 \cong S^{21}.$$

Main theorem package

Main theorem package. Let Ω' be the row-graded parameter space on $M = S \times S$ satisfying left-row output, the fixed same-row off-diagonal Steiner rule, and σ-equivariance. Then:

1. The parameter space has normal form

$$\Omega' \cong S^{18} \times S^3 \cong S^{21}, \qquad |\Omega'| = 3^{21} = 10\,460\,353\,203.$$

2. The associativity spectrum satisfies

$$63 \leq \mathrm{Assoc} \leq 597,$$

with exact endpoint loci and partition function. The path-resolved Hamming defect satisfies

$$\min H_{tot} = -2268, \qquad \max H_{tot} = 7302, \qquad \max_{N_- = 0} H_{tot} = 7020.$$

3. The four internal selection conditions

$$\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad (r,r)^2 = (r,r), \qquad \mathrm{StabMaps}_R = \{C, J\}$$

select the right-row projection magma RP uniquely, with multiplication

$$(r_1, c_1) \cdot (r_2, c_2) = \begin{cases} (r_1, r_2), & r_1 \neq r_2, \\ (r_1, \overline{c_1 c_2}), & r_1 = r_2,\ c_1 \neq c_2, \\ (r_1, r_1), & r_1 = r_2,\ c_1 = c_2. \end{cases}$$

4. The selected magma has the post-selection structural profile proved in Part III: left translations and left-translation semigroup, endomorphisms and intertwiners, binary term operations, matrix row-sum contraction, an operator interpretation of the H-statistic, associated compact Lie groups, and a signed symplectic lift.

5. The finite distribution and profile claims are supported by the certificate protocol recorded in the companion material.

Proof. The normal form is Theorem 1.1. The associativity spectrum is Theorem 3.5, with endpoint and distribution data in Theorems 3.6 and 3.7. The path-resolved Hamming defect is Theorem 4.2. The selection statement is Theorem 8.1. The post-selection structural profile is given by Lemma 9.1, Theorems 10.2, 10.3, 10.5, 10.7, 11.2, and 12.3, together with Propositions 10.6, 11.3, 11.4, and 13.1, and Theorems 14.1, 15.1, and 16.1. The certificate interface is recorded in Section A. ☐

Proof architecture

The proof separates the roles of the global invariants and selection conditions.

component	role
Part I: finite parameter space	normal form, effective symmetry, associativity spectrum and distribution, and path-resolved Hamming defect
Part II: intrinsic selection	four selection conditions and the uniqueness proof for RP
Part III: linearization	algebraic and realization-level structures recorded after RP is fixed
Companion material	finite certificate protocol, RP one-page structural profile, structural reading, open problems, and assistance note

Associativity and the path-resolved Hamming defect are global invariants. The four conditions in the main theorem package are selection conditions. The compact Lie groups appearing in Part III are compact Lie groups associated with specified linear structures on the shared carrier $k[M] \cong \mathrm{Mat}_3(k)$.

Guide to the paper

Part I constructs the finite parameter space and records its two global invariants. Part II applies the internal selection conditions and proves uniqueness of RP. Part III fixes the selected multiplication and develops its operator, term-operation, endomorphism, matrix, compact Lie group, and symplectic structures. The companion material records the certificate path, a compact structural profile for the selected point RP, the row-complement comparison, the Landauer/Shannon structural reading, and open problems.

Part I

The finite parameter space

Part I constructs the ambient finite space and records the two global invariants used later: associativity and the path-resolved Hamming defect. The statements here describe the parameter space before selection is applied.

1 Carrier, axioms, and normal form

1.1 Carrier and shift

Let
$$S = \mathbb{F}_3, \qquad M = S \times S.$$
All arithmetic in row and column coordinates is in S. Equivalently, $M = \mathbb{F}_3^2$ is the point set of the affine plane $\mathrm{AG}(2,3)$, with the chosen row–column coordinates fixed throughout. For $x = (r,c) \in M$, write
$$\mathrm{row}(x) = r, \qquad \mathrm{col}(x) = c,$$
and let
$$F_r = \{r\} \times S$$
be the row fiber. We use the cyclic shift
$$\sigma(r,c) = (r+1, c+1).$$
Thus σ translates rows and columns simultaneously.

The row-relative offset $c - r$ is invariant under σ. Hence M decomposes into the three σ-orbits
$$D_i = \{(r, r+i) : r \in S\}, \qquad i \in S.$$
The orbit D_0 is the diagonal orbit, while D_1 and D_2 are the two non-diagonal orbits.

For distinct $i, j \in S$, write $\overline{ij}$ for the third element of $S \setminus \{i,j\}$. Equivalently, in $\mathbb{F}_3$,
$$\overline{ij} = -i - j \qquad (i \neq j).$$

1.2 Axioms

The parameter space Ω' consists of binary operations $\cdot$ on M satisfying the following three conditions.

AX0, left-row output.

For all $x, y \in M$,
$$\mathrm{row}(x \cdot y) = \mathrm{row}(x).$$
Equivalently, $x \cdot y \in F_{\mathrm{row}(x)}$.

σ-equivariance.

For all $x, y \in M$,
$$\sigma(x \cdot y) = \sigma(x) \cdot \sigma(y).$$

AX2, same-row off-diagonal Steiner rule.

For all $r, i, j \in S$ with $i \neq j$,
$$(r, r+i) \cdot (r, r+j) = (r, r + \overline{ij}).$$
Equivalently, in absolute column notation,
$$(r, c_1) \cdot (r, c_2) = (r, \overline{c_1 c_2}) \qquad (c_1 \neq c_2).$$
This is the Steiner multiplication on a three-point row fiber [23, 9].

AX0 fixes only the output row. The output column remains free in the cross-row branch and on the same-point diagonal branch.

1.3 Normal coordinates

It is convenient to use row-relative coordinates adapted to this orbit decomposition. A point of row r in D_i is written as $(r, r+i)$, where $i \in S$ is the column offset from the row. Thus i is the σ-orbit label.

A normal coordinate is a triple

$$(A, B, d) \in S^9 \times S^9 \times S^3,$$

where

$$A = (A_{ij})_{i,j \in S}, \qquad B = (B_{ij})_{i,j \in S}, \qquad d = (d_0, d_1, d_2).$$

The associated multiplication is defined by the four formulas

$$(r, r+i) \cdot (r+1, r+1+j) = (r, r+A_{ij}),$$

$$(r, r+i) \cdot (r+2, r+2+j) = (r, r+B_{ij}),$$

$$(r, r+i) \cdot (r, r+i) = (r, r+d_i),$$

and, for $i \neq j$,

$$(r, r+i) \cdot (r, r+j) = (r, r+\overline{ij}).$$

The first two arrays record the two ordered cross-row representatives $(0, 1)$ and $(0, 2)$. The vector d records the same-point branch by σ-orbit label: d_i is the output offset of x^2 for $x \in D_i$. The fourth formula is AX2.

Conversely, any operation satisfying AX0, σ-equivariance, and AX2 has a unique coordinate triple (A, B, d): after applying a power of σ, every cross-row product reduces to one of the two row-pair representatives above, and every same-point product reduces to one of the three diagonal coordinates.

1.4 Normal form theorem

Theorem 1.1. The AX0+AX2, σ-equivariant parameter space has normal form

$$\Omega' \cong S^{18} \times S^3 \cong S^{21}.$$

Consequently

$$|\Omega'| = 3^{21} = 10\,460\,353\,203.$$

Proof. AX0 fixes the output row of every product. AX2 fixes all same-row off-diagonal output columns. The same-point diagonal branch has three row-relative coordinates d_0, d_1, d_2. For cross-row products, σ-equivariance reduces the ordered row-pair data to the two representatives $(0, 1)$ and $(0, 2)$. Each representative has 3×3 column-offset inputs, hence contributes nine independent coordinates. Thus the cross-row branch contributes 18 coordinates and the diagonal branch contributes 3 coordinates.

The coordinate construction above gives a product satisfying AX0, σ-equivariance, and AX2 for every $(A, B, d) \in S^9 \times S^9 \times S^3$, and the preceding paragraph recovers these coordinates uniquely from any such product. Hence $\Omega' \cong S^{21}$, and its cardinality is 3^{21}. $\square$

2 Effective symmetry

The oriented row-graded setup fixes the generator $s \mapsto s+1$ through $\sigma(r,c) = (r+1, c+1)$. For parameter-space compression we use the normalizer of this cyclic shift in $\mathrm{Sym}(S)$. Translations $s \mapsto s+\beta$ commute with the shift and act trivially on the row-relative normal coordinates. The remaining normalizer class is represented by the reflection $\nu(s) = -s$, which conjugates σ to σ^{-1}. Since σ-equivariance is equivalent to σ^{-1}-equivariance, this reflection preserves Ω' as a parameter-space compression symmetry.

On normal coordinates it induces the involution

$$\iota(A, B, d) = (A', B', d')$$

with indices in S and

$$A'_{ij} = -B_{-i,-j}, \qquad B'_{ij} = -A_{-i,-j}, \qquad d'_i = -d_{-i}.$$

We denote the resulting residual effective action by

$$G_{\mathrm{eff}} = \langle \iota \rangle \cong C_2.$$

On the diagonal coordinate sector $\Delta = S^3$, the same residual reflection gives the compression

$$|\Delta/G_{\mathrm{eff}}| = \frac{3^3+3}{2} = 15.$$

Proposition 2.1. The effective orbit count of the full normal-coordinate parameter space is

$$|\Omega'/G_{\mathrm{eff}}| = \frac{3^{21}+3^{10}}{2}.$$

Proof. Burnside's lemma applies to the displayed involution. The identity contributes 3^{21}. For a fixed point of ι, the nine entries of A are free and then determine B by $B_{ij} = -A_{-i,-j}$. The diagonal equations are

$$d_0 = -d_0, \qquad d_1 = -d_2, \qquad d_2 = -d_1,$$

so $d_0 = 0$ and d_1 is free. Hence the fixed locus has $9+1 = 10$ ternary degrees of freedom and size 3^{10}. Therefore

$$|\Omega'/G_{\mathrm{eff}}| = \frac{3^{21}+3^{10}}{2}.$$

□

The orbit count is a parameter-space compression. The residual reflection records the finite normal-coordinate symmetry available at the parameter-space level. The directed-edge geometry used later is formulated on $M^{\times}$ itself.

3 Associativity spectrum and distribution

3.1 Block decomposition

The associativity statistic is a global invariant on Ω'. For $x \in \Omega'$, define

$$\mathrm{Assoc}(x) = \#\{(u,v,w) \in M^3 : (u\cdot v)\cdot w = u\cdot(v\cdot w)\}.$$

Since $|M| = 9$, this statistic lies between 0 and 729.

Theorem 3.1. In normal coordinates,

$$\mathrm{Assoc} = 3\,A_{\mathrm{norm}},$$

where A_{norm} is a normalized sum over $3^5 = 243$ triples and decomposes as

$$A_{\mathrm{norm}} = A_{RRR} + A_{RRS} + A_{RSR} + A_{SRR} + A_{\mathrm{dist}}.$$

The five blocks correspond to the row patterns

$$RRR, \qquad RRS, \qquad RSR, \qquad SRR, \qquad DIST.$$

Proof. By σ-equivariance the first row coordinate of a triple may be normalized to 0. The two remaining row variables and the three column variables give 3^5 normalized cases; the factor 3 restores the three translates. The displayed blocks are the five equality patterns of the three rows. □

3.2 Same-row block

The all-same-row block depends only on the diagonal vector d. Let

$$E(d) = \#\{(i,j) : i \neq j,\ d_i = d_j\}, \qquad N(d) = \#\{(i,j) : i \neq j,\ d_i = j,\ d_j = j\}.$$

Lemma 3.2. The same-row contribution is

$$A_{RRR}(d) = 9 + E(d) + 2N(d).$$

The four diagonal classes give normalized same-row values

$$9, \qquad 11, \qquad 13, \qquad 19.$$

In particular,

$$A_{RRR}(000) = 19.$$

Proof. In one row, unequal column pairs use the Steiner rule and equal pairs use d. The 27 triples split into a baseline family plus the two diagonal-sensitive families counted by $E(d)$ and $N(d)$. This gives the formula and the four class values. □

3.3 Column-independent profile at $d = 000$

A cross-row rule is column-independent when its output column depends only on the two row coordinates. By σ-equivariance such a rule is determined by two anchors $a = h(0,1)$ and $b = h(0,2)$. Let $h_{a,b}$ be the column-independent rule

$$h(r, r+1) = r + a, \qquad h(r, r+2) = r + b.$$

At $d = 000$, same-point products satisfy $(r,c)^2 = (r,r)$.

Proposition 3.3. On the column-independent, $d = 000$ stratum,

$$\begin{aligned}(A_{RRR}, A_{RRS}, A_{RSR}, A_{SRR}, A_{\mathrm{dist}})(a,b,000)\\ = \big(19,\ 9[a=0] + 9[b=0],\ 9[a=0] + 9[b=0],\\ 54,\ 54[a=b]\big).\end{aligned}$$

Consequently

$$\boxed{\mathrm{Assoc}_{000}(h_{a,b}) = 3\,(73 + 18[a=0] + 18[b=0] + 54[a=b])\,.}$$

Proof. The RRR value is Lemma 3.2. In the RRS and RSR patterns, the same-row step must return the cross-row anchor; at $d = 000$ this contributes 9 exactly when the relevant anchor is 0. In SRR, the final cross-row output ignores same-row column processing, giving all 54 normalized triples. In $DIST$, the two association orders compare the two cross-row anchors, so all 54 triples survive exactly when $a = b$. □

The nine column-independent rules split as follows:

class	representatives	profile	Assoc_{000}
anchored trivial	$(0,0)$	$(19,18,18,54,54)$	489
off-anchor trivial	$(1,1),(2,2)$	$(19,0,0,54,54)$	381
partly anchored nondegenerate	$(0,1),(0,2),(1,0),(2,0)$	$(19,9,9,54,0)$	273
fully off-anchor nondegenerate	$(1,2),(2,1)$	$(19,0,0,54,0)$	219

The fourth class consists of the selected point $\mathsf{RP} = (a,b,d) = (1,2,000)$ and the row-complementary point $\mathsf{RC} = (a,b,d) = (2,1,000)$.

3.4 Global range

Lemma 3.4 (Associativity spectrum and distribution certificate protocol)**.** The global associativity spectrum and distribution is used through five finite obligations.

1. **A1, block reduction.** The associativity statistic is reduced by σ-normalization to the five row-pattern blocks
$$A_{\mathrm{norm}} = A_{RRR} + A_{RRS} + A_{RSR} + A_{SRR} + A_{\mathrm{dist}}$$
of Theorem 3.1.
2. **A2, diagonal decomposition.** The same-row block is evaluated from the diagonal vector by Lemma 3.2, giving the four possible values
$$9, \quad 11, \quad 13, \quad 19.$$
3. **A3, fixed-diagonal cross-range certificates.** For each diagonal class, the certificate verifies the exact range of
$$A_{\mathrm{cross}} = A_{RRS} + A_{RSR} + A_{SRR} + A_{\mathrm{dist}}.$$
4. **A4, endpoint witnesses and endpoint counts.** The endpoint certificates verify the normal-coordinate witnesses, endpoint counts, and orbit representatives for $\mathrm{Assoc} = 63$ and $\mathrm{Assoc} = 597$.
5. **A5, distribution consistency.** The partition-function certificate gives the full coefficient table for
$$Z(q) = \sum_{x \in \Omega'} q^{\mathrm{Assoc}(x)}$$
and checks total mass, range, moments, and mode.

Any certificate satisfying A1–A5 proves the global associativity range, endpoint loci, and partition-function claims in this spectrum-and-distribution section. □

Theorem 3.5 (Global associativity spectrum and distribution)**.** On Ω',

$$21 \leq A_{\mathrm{norm}} \leq 199.$$

Equivalently,

$$63 \leq \mathrm{Assoc} \leq 597.$$

The finite certificate is organized by diagonal class:

diagonal class	A_{RRR}	certified A_{cross} range	A_{norm} consequence
permutation	9	12..180	21..189
two-valued, no fixed point	11	10..162	21..173
two-valued, with fixed point	13	12..174	25..187
constant	19	12..180	31..199

Here

$$A_{\mathrm{cross}} = A_{RRS} + A_{RSR} + A_{SRR} + A_{\mathrm{dist}}.$$

The endpoint values are therefore $A_{\mathrm{norm}} = 21$ and $A_{\mathrm{norm}} = 199$, or $\mathrm{Assoc} = 63$ and $\mathrm{Assoc} = 597$.

Proof. The hand-readable reduction is Theorem 3.1 plus Lemma 3.2. After that reduction, the global claim is exactly the A3 cross-range obligation of Lemma 3.4. Adding the certified cross-row ranges to the four possible same-row values gives the displayed global range.

The endpoint witnesses, endpoint counts, orbit representatives, and partition-function consistency are the A4–A5 obligations of Lemma 3.4. They are recorded in the `L1/tables/` distribution files listed in `SUPPORT_INDEX.txt` and checked by `L1/scripts/verify_layer1_v3_final.py`. ☐

Spectrum landmarks. The selected point has

$$\mathrm{Assoc}(\mathsf{RP}) = 219,$$

while the parameter-space minimum is 63. On the column-independent $d = 000$ profile, the value $\mathrm{Assoc}_{000} = 219$ is attained by the two fully off-anchor nondegenerate rules RP and RC.

3.5 Endpoint loci and partition function

3.5.1 Endpoint loci

Theorem 3.6. The associativity endpoint loci have sizes

$$|\mathcal{M}_{63}| = 24, \qquad |\mathcal{M}_{63}/\,\mathrm{Sym}(S)| = 12,$$

and

$$|\mathcal{M}_{597}| = 6, \qquad |\mathcal{M}_{597}/\,\mathrm{Sym}(S)| = 3.$$

The minimum endpoint block profiles are

$$(11, 2, 2, 2, 4) \quad \text{and} \quad (9, 2, 2, 4, 4),$$

while the maximum endpoint block profile is

$$(19, 36, 36, 54, 54).$$

Proof. This is the A4 endpoint part of Lemma 3.4. The endpoint certificates record the exact solution counts by diagonal sector, orbit representatives, and explicit witnesses in normal coordinates. The minimum occurs in the permutation and two-valued/no-fixed diagonal classes. The maximum occurs in the constant diagonal class. The block profiles are obtained by evaluating the five normalized associativity blocks on the endpoint witnesses. □

3.5.2 Exact associativity partition function

Theorem 3.7. The exact polynomial

$$Z(q) = \sum_{x \in \Omega'} q^{\mathrm{Assoc}(x)}$$

has 167 nonzero coefficients and total mass

$$Z(1) = 3^{21}.$$

Its first two moments are

$$\mathbb{E}[\mathrm{Assoc}] = 245, \qquad \mathrm{Var}(\mathrm{Assoc}) = \frac{27820}{27}.$$

The mode occurs at

$$\mathrm{Assoc} = 237$$

with coefficient

$$426{,}317{,}976.$$

Proof. This is the A5 partition-function part of Lemma 3.4. The coefficient certificate in `L1/tables/` is the exact finite distribution of Assoc on Ω'. The total mass, range, mean, variance, and mode are computed directly from that table and checked by `L1/scripts/verify_layer1_v3_final.py`. □

4 Path-resolved Hamming defect

4.1 Local distance reduction

The path-resolved Hamming-defect statistic is the second global invariant. It is a finite path-information defect. The continuation side keeps the two-step route data, while the left-translation side compares only the collapsed endpoint operators. Thus $h_q/2$ is the signed Hamming contrast between a path-resolved comparison and an endpoint-resolved comparison. Negative local values are precisely the local cancellation failures counted by N_-. The normalization $H_{tot} = 6H_{\mathrm{norm}}$ is the σ-orbit normalization used by the path-resolved Hamming defect certificate.

Its local index is

$$q = (b, t, a, e, f) \in S^5.$$

For a completed normal-form point $x \in \Omega'$, let M_s denote the local table read: M_1 and M_2 are the two cross-row tables, and M_0 is the same-row diagonal/complement rule. All row and column arithmetic below is in $S = \mathbb{F}_3$. Define

$$xy = M_b(a, e), \qquad yz = M_{t-b}(e - b, f - b), \qquad z_R = b + yz,$$

and

$$ep_L = M_t(xy, f), \qquad ep_R = M_b(a, z_R).$$

For $c \in S$, define the collapsed left-translation signature

$$L_c(i, j) = M_i(c, j) \qquad ((i, j) \in S^2).$$

For $s, \alpha, z \in S$, define the stepwise continuation signature

$$C_{s,\alpha,z}(i, j) = M_s(\alpha,\ s + M_{i-s}(z - s, j - s)) \qquad ((i, j) \in S^2).$$

The associated Hamming distances are

$$D_L(c, c') = \#\{(i, j) \in S^2 : L_c(i, j) \neq L_{c'}(i, j)\},$$

and

$$D_C(s, \alpha, z; s', \alpha', z') = \#\{(i, j) \in S^2 : C_{s,\alpha,z}(i, j) \neq C_{s',\alpha',z'}(i, j)\}.$$

Direct local definition. For each of the nine test pairs $(i, j) \in S^2$, set

$$\delta^C_{ij}(q; x) = \mathbf{1}\{C_{t,xy,f}(i, j) \neq C_{b,a,z_R}(i, j)\},$$

and

$$\delta^L_{ij}(q; x) = \mathbf{1}\{L_{ep_L}(i, j) \neq L_{ep_R}(i, j)\}.$$

The nine direct local summands are

$$\Delta_{ij}(q; x) = 2\big(\delta^C_{ij}(q; x) - \delta^L_{ij}(q; x)\big),$$

displayed as

	$j = 0$	$j = 1$	$j = 2$
$i = 0$	Δ_{00}	Δ_{01}	Δ_{02}
$i = 1$	Δ_{10}	Δ_{11}	Δ_{12}
$i = 2$	Δ_{20}	Δ_{21}	Δ_{22}

and the direct Hamming-defect contribution is

$$h_q(x) = \sum_{(i,j) \in S^2} \Delta_{ij}(q; x).$$

Theorem 4.1 (Local distance reduction)**.** For every $x \in \Omega'$ and $q \in S^5$,

$$\boxed{h_q(x) = 2\left(D_C(t, xy, f; b, a, z_R) - D_L(ep_L, ep_R)\right).}$$

Consequently the local direct definition and the Hamming-distance formula define the same global observables H_{norm} and H_{tot}.

Proof. By definition, $D_C(t, xy, f; b, a, z_R)$ is the sum of the nine indicators $\delta^C_{ij}(q; x)$, and $D_L(ep_L, ep_R)$ is the sum of the nine indicators $\delta^L_{ij}(q; x)$. Subtracting these two Hamming distances and multiplying by 2 gives exactly the displayed sum of the nine direct local summands Δ_{ij}. The symbolic reduction is recorded by the path-resolved Hamming defect certificate handles `HF-RED` and `HF-CERT-IF`. □

Then

$$H_{\text{norm}}(x) = \sum_{q \in S^5} \frac{h_q(x)}{2}, \qquad H_{tot}(x) = 6\, H_{\text{norm}}(x) = 3 \sum_{q \in S^5} h_q(x),$$

and

$$\boxed{N_-(x) = 3\,\#\{q \in S^5 : h_q(x) < 0\}.}$$

Thus the nonnegative local condition is the finite system

$$N_-(x) = 0 \quad \Longleftrightarrow \quad h_q(x) \geq 0 \quad (q \in S^5).$$

For the signed-layer diagnostics, set

$$H_{pos}(x) = 3 \sum_{q \in S^5} \max\{h_q(x), 0\}, \qquad H_{\text{neg\_abs}}(x) = 3 \sum_{q \in S^5} \max\{-h_q(x), 0\}.$$

Then

$$H_{tot}(x) = H_{pos}(x) - H_{\text{neg\_abs}}(x).$$

Also write

$$h_{\text{loc,min}}(x) = \min_{q \in S^5} h_q(x), \qquad h_{\text{loc,max}}(x) = \max_{q \in S^5} h_q(x).$$

4.2 Global range theorem

The global range theorem for the path-resolved Hamming defect is supported by four finite certificate obligations.

handle	obligation	result
H1	unrestricted full-parameter-space range	$H_{\text{norm}} \in [-378, 1217]$, endpoint counts $8, 12$
H2	nonnegative-local-locus exclusion	$(h_q \geq 0\ \forall q) \wedge H_{\text{norm}} \geq 1171$ is UNSAT
H3	nonnegative-local-locus witnesses	RP and RC have $H_{\text{norm}} = 1170,\ N_- = 0$
H4	maximal layer with negative local contributions	exactly twelve points above the nonnegative local locus, all with $(H_{\text{norm}}, N_-) = (1217, 3)$

Theorem 4.2 (Global range of the path-resolved Hamming defect)**.** On $\Omega' = S^{21}$:

1. The unrestricted range is

   $$\boxed{\min H_{tot} = -2268, \qquad \max H_{tot} = 7302.}$$

   Equivalently, $H_{\text{norm}} \in [-378, 1217]$. The endpoint multiplicities are 8 and 12.

2. The nonnegative local locus is

   $$\boxed{\max\{H_{tot} : N_- = 0\} = 7020.}$$

3. The region above the nonnegative local locus is exactly the unrestricted maximum layer:

$$\boxed{\{H_{tot} > 7020\} = \{H_{tot} = 7302\}.}$$

This layer has 12 points, all with

$$H_{\text{norm}} = 1217, \qquad H_{tot} = 7302, \qquad N_{-} = 3,$$

$$H_{pos} = 7308, \qquad H_{\text{neg\_abs}} = 6, \qquad h_{\text{loc,min}} = -2, \qquad h_{\text{loc,max}} = 18.$$

Proof. The analytic reduction is Theorem 4.1. H1 covers all 3^{21} normal-form words and gives the unrestricted range. For the nonnegative local locus, $N_{-} = 0$ is the system $h_q \geq 0$ for all $q \in S^5$, and $H_{tot} > 7020$ is equivalent to $H_{\text{norm}} \geq 1171$. H2 proves

$$(h_q \geq 0 \ \forall q \in S^5) \wedge H_{\text{norm}} \geq 1171$$

UNSAT on all 18540 depth-9 live nodes:

$$18540/18540 \text{ UNSAT}, \qquad \text{SAT} = 0, \qquad \text{UNKNOWN} = 0.$$

H3 supplies nonnegative-local witnesses at $H_{\text{norm}} = 1170$, and H4 classifies the full layer above the nonnegative-local threshold as the twelve maximizers with negative local contributions. □

4.3 Nonnegative local locus and maximal signed layer

Corollary 4.3. RP and RC satisfy

$$H_{tot} = 7020, \qquad H_{\text{norm}} = 1170, \qquad N_{-} = 0.$$

Thus both lie on the nonnegative local maximum of the path-resolved Hamming defect.

Proof. Direct evaluation of

$$\mathsf{RP} = 111111111222222222000, \qquad \mathsf{RC} = 222222222111111111000$$

gives the displayed values, and Theorem 4.2 proves that no point in the nonnegative local locus has larger H_{tot}. □

Part II

Intrinsic finite selection

Part I gives the finite invariant data for the ambient parameter space. Part II defines the selection conditions and proves that their unique survivor is RP. The first two conditions are compression conditions, the third fixes the compressed diagonal convention, and the fourth is the directed-edge geometry condition.

5 Selection conditions

For a normal-form point of Ω', write g for its cross-row output column rule and $d = (d_0, d_1, d_2)$ for its same-point diagonal. The selection conditions are the conjunction

$$\boxed{\mathcal{H}_{acc}(g) = 0, \qquad H_{diag}(d) = 0, \qquad (r,r)^2 = (r,r)\ (r \in S), \qquad \mathrm{StabMaps}_R = \{C, J\}.}$$

The first condition is zero conditional mutual information. The second condition is zero diagonal entropy. The third fixes the diagonal-orbit representative after entropy compression; equivalently, it anchors idempotence on the diagonal orbit D_0. The fourth chooses the intrinsic head-row continuation/reversal geometry on the six directed edges.

The first two conditions are information-theoretic: $\mathcal{H}_{acc}$ is conditional mutual information and H_{diag} is Shannon entropy [26, 10]. The Landauer–Bennett reading is the implementation-independent economy reading of avoidable column-coordinate dependence and discarded diagonal information [19, 3, 4].

6 Conditional independence and zero diagonal entropy

6.1 Column-independent cross-row rules

For $r_1 \neq r_2$, write the cross-row branch as

$$(r_1, c_1) \cdot (r_2, c_2) = (r_1, g(r_1, c_1, r_2, c_2)).$$

A cross-row rule is *column-independent* if the output column depends only on its ordered row pair:

$$g(r_1, c_1, r_2, c_2) = h(r_1, r_2).$$

Equivalently, after (r_1, r_2) is fixed, the output column is constant over the nine choices of (c_1, c_2).

By σ-equivariance, a column-independent rule is determined by two anchors

$$a = h(0, 1), \qquad b = h(0, 2),$$

and the remaining ordered row pairs satisfy

$$h(r, r+1) = r + a, \qquad h(r, r+2) = r + b.$$

There are $3^2 = 9$ column-independent cross-row rules; denote them by $h_{a,b}$.

6.2 Conditional mutual information

Let

$$\Sigma_{cross} = \{(r_1, c_1, r_2, c_2) \in S^4 : r_1 \neq r_2\}$$

with the uniform distribution, and let

$$C_{out} = g(R_1, C_1, R_2, C_2)$$

be the output-column random variable. Define

$$\boxed{\mathcal{H}_{acc}(g) = I(C_{out}; C_1, C_2 \mid R_1, R_2).}$$

Logarithms are base 2.

Theorem 6.1 (Conditional-independence criterion)**.** For a cross-row rule g,

$$\mathcal{H}_{acc}(g) = 0 \iff g(r_1, c_1, r_2, c_2) = h(r_1, r_2).$$

Thus the zero conditional mutual information locus consists of the nine column-independent rules $h_{a,b}$.

Proof. Conditional mutual information vanishes exactly when

$$C_{out} \perp (C_1, C_2) \mid (R_1, R_2).$$

Since C_{out} is a deterministic function of (R_1, C_1, R_2, C_2), this conditional independence is equivalent to constancy in (C_1, C_2) for every fixed ordered row pair. This is exactly column-independence. The ordered row-pair representatives $(0, 1)$ and $(0, 2)$ then supply the two anchors $(a, b) \in S^2$. □

6.3 Diagonal entropy

For a diagonal triple $d = (d_0, d_1, d_2) \in S^3$, let

$$p_s = \frac{1}{3}\#\{i : d_i = s\}.$$

The diagonal Shannon observable is

$$\boxed{H_{diag}(d) = -\sum_{s \in S} p_s \log_2 p_s.}$$

The possible levels are

diagonal type	multiplicities	H_{diag}
constant	$(3, 0, 0)$	0
two-valued	$(2, 1, 0)$	$\log_2 3 - \frac{2}{3}$
permutation	$(1, 1, 1)$	$\log_2 3$

Theorem 6.2 (Zero diagonal entropy)**.** For diagonal triples,

$$H_{diag}(d) = 0 \iff d_0 = d_1 = d_2.$$

Thus zero diagonal entropy reduces the diagonal sector from 27 choices to

$$000, \qquad 111, \qquad 222.$$

Together with zero conditional mutual information, the selection domain becomes 9×3.

Proof. Entropy of a finite distribution vanishes exactly when the distribution is concentrated on one symbol. For the empirical distribution of (d_0, d_1, d_2), this is precisely constancy of the diagonal triple. □

6.4 Idempotence on the diagonal orbit

The diagonal orbit D_0 consists of the points (r, r).

Proposition 6.3 (Idempotence on the diagonal orbit)**.** After zero diagonal entropy, the condition

$$(r, r) \cdot (r, r) = (r, r) \qquad (r \in S)$$

selects the unique constant diagonal

$$d = 000.$$

For this diagonal,

$$(r, r + i)^2 = (r, r) \qquad (r, i \in S),$$

so the idempotent points are exactly the diagonal orbit D_0.

Proof. On the compressed diagonal sector $d = sss$, the equation $(r, r)^2 = (r, r)$ gives $s = 0$. Hence the unique compressed diagonal compatible with idempotence on the diagonal orbit is 000. The displayed square formula follows by substituting $d = 000$ into the same-point branch. □

The first three selection conditions therefore reduce the normal-form coordinates to

$$h_{a,b}, \qquad (a, b) \in S^2, \qquad d = 000.$$

7 Directed-edge geometry

7.1 The maps C and J

Let

$$M^\times = \{(r, c) \in M : r \neq c\}$$

be the six-point directed-edge set of the triangle on vertex set S. In the orbit notation above,

$$M^\times = D_1 \sqcup D_2.$$

The point (r, c) is the directed edge with tail r and head c. Define

$$C(r, c) = (c, \overline{rc}), \qquad J(r, c) = (c, r).$$

The map C advances the edge through the third vertex, while J reverses it.

Theorem 7.1 (Directed-edge dihedral relations)**.** On $M^\times$,

$$C^3 = 1, \qquad J^2 = 1, \qquad JCJ = C^{-1}.$$

Thus C and J generate the dihedral geometry of the directed triangle.

Proof. Applying C three times gives

$$(r, c) \mapsto (c, \overline{rc}) \mapsto (\overline{rc}, r) \mapsto (r, c).$$

Applying J twice reverses the edge twice. Finally,

$$JCJ(r, c) = JC(c, r) = J(r, \overline{rc}) = (\overline{rc}, r) = C^{-1}(r, c).$$

□

For every relabelling $\rho \in \mathrm{Sym}(S)$, acting by $\rho_\times(r,c) = (\rho(r), \rho(c))$, one has

$$\rho_\times C = C\rho_\times, \qquad \rho_\times J = J\rho_\times.$$

The pair $\{C, J\}$ is the intrinsic head-row pair:

$$\{C(r,c), J(r,c)\} = \{y \in M^\times : \mathrm{tail}(y) = c\}.$$

7.2 Right stabilizer maps

For $x \in M^\times$, define

$$\mathrm{Stab}_R(x) = \{y \in M^\times \setminus \{x\} : x \cdot y = x\}.$$

On the anchored selection domain $h_{a,b}$, $d = 000$, right-stabilization of $x = (r,c)$ is governed by the cross-row equation

$$h(r, r_y) = c.$$

The C/J right-stabilizer condition is the finite condition

$$\mathrm{Stab}_R(x) = \{C(x), J(x)\} \qquad (x \in M^\times).$$

Equivalently, the anchored rule has two right-stabilizer maps, and they form

$$\mathrm{StabMaps}_R = \{C, J\}.$$

7.3 C/J right-stabilizer condition

Theorem 7.2 (C/J right-stabilizer condition on the anchored domain)**.** For a column-independent rule $h_{a,b}$ with $d = 000$,

$$\mathrm{StabMaps}_R = \{C, J\} \quad \Longleftrightarrow \quad (a,b) = (1,2).$$

The corresponding cross-row branch is the right-row projection branch.

Proof. Let $x = (r,c) \in M^\times$, and write $\delta = c - r \in \{1, 2\}$. For the two cross rows of a potential right stabilizer, the right-stabilizer equation becomes

$$r_y = r + 1 \Rightarrow a = \delta, \qquad r_y = r + 2 \Rightarrow b = \delta.$$

The C/J right-stabilizer condition requires the right-stabilizer neighbours of (r,c) to be exactly

$$J(r,c) = (c,r), \qquad C(r,c) = (c, \overline{rc}),$$

so both right-stabilizer neighbours must lie in row c.

For $\delta = 1$, this forces right-stabilization in row $r + 1 = c$, hence $a = 1$, and excludes right-stabilization in row $r + 2$, hence $b \neq 1$. For $\delta = 2$, it forces right-stabilization in row $r + 2 = c$, hence $b = 2$, and excludes right-stabilization in row $r + 1$, hence $a \neq 2$. Therefore $(a,b) = (1,2)$.

Conversely, if $(a,b) = (1,2)$, then for every $x = (r,c)$ the only right-stabilizing cross row is row c, and the two off-diagonal points in that row are (c,r) and $(c, \overline{rc})$, namely $J(x)$ and $C(x)$. Thus $\mathrm{StabMaps}_R = \{C, J\}$. $\square$

8 Uniqueness theorem

Theorem 8.1 (Uniqueness theorem)**.** The four conditions

$$\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad (r,r)^2 = (r,r)\ (r \in S), \qquad \mathrm{StabMaps}_R = \{C, J\}$$

select RP uniquely inside Ω'.

Proof. The conditional-independence criterion gives a column-independent rule $h_{a,b}$ by Theorem 6.1. Zero diagonal entropy gives a constant diagonal by Theorem 6.2, and idempotence on the diagonal orbit fixes it to $d = 000$ by Proposition 6.3. On this anchored domain, the C/J right-stabilizer condition is equivalent to $(a,b) = (1,2)$ by Theorem 7.2. This is exactly the right-row projection multiplication, with

$$(r_1, c_1) \cdot (r_2, c_2) = (r_1, r_2) \qquad (r_1 \neq r_2),$$

and same-point diagonal $d = 000$. Hence the survivor is unique and is RP. $\square$

Part III

Linearization of the selected magma

Part III fixes the selected multiplication RP and records the structures that appear after this choice. The first layer is algebraic: the leftmost-variable row law, left translations, left-translation semigroup, row-fiber algebra, endomorphisms, intertwiners, and binary term operations. The second layer is realization-level: row-sum contraction on $k[M] \cong \mathrm{Mat}_3(k)$, the operator interpretation of the H-statistic, associated compact Lie groups, and the signed symplectic lift of the C/J right-stabilizer geometry.

9 The selected operation

9.1 Multiplication and ground-field conventions

The selected multiplication is

$$(r_1, c_1) \cdot (r_2, c_2) = \begin{cases} (r_1, r_2), & r_1 \neq r_2, \\ (r_1, \overline{c_1 c_2}), & r_1 = r_2,\ c_1 \neq c_2, \\ (r_1, r_1), & r_1 = r_2,\ c_1 = c_2. \end{cases}$$

The first case is the right-row projection branch; the second is the fixed same-row Steiner branch; the third is diagonal anchoring.

The finite magma statements are set-theoretic. When the carrier is linearized, let k be a ground field and let $k[M]$ be the vector space with basis e_x, $x \in M$. Rank and image statements for individual finite maps are valid over any ground field. Linear span statements are over the chosen k. Spectral, Jordan, and fiber-algebra statements involving $t \pm 1$ or idempotents with coefficients $1/2$ are stated under

$$\operatorname{char} k \neq 2.$$

The associated compact Lie group statements and signed symplectic realization statements are over $\mathbb{R}$ unless stated otherwise.

9.2 Leftmost-variable row law

For a nonempty term $t(x_1, \ldots, x_n)$, let $\ell(t)$ be the leftmost variable appearing in the term tree.

Lemma 9.1 (Leftmost-variable row lemma)**.** For every RP term $t(x_1, \ldots, x_n)$,

$$\operatorname{row}\big(t(x_1, \ldots, x_n)\big) = \operatorname{row}\big(\ell(t)\big).$$

Proof. The assertion is immediate for a variable. If $t = u \cdot v$, then the left-row output axiom gives $\operatorname{row}(t) = \operatorname{row}(u \cdot v) = \operatorname{row}(u)$. The induction hypothesis identifies this row with the row of the leftmost variable of u, which is also the leftmost variable of $u \cdot v$. ☐

The leftmost-variable row lemma is the common source of the row-target product law for left-translation compositions and the leftmost-variable coordinate in the binary term operations.

10 Left translations and generated semigroup

For $x \in M$, write $L_x(y) = x \cdot y$.

10.1 Rank and image

Proposition 10.1. For $x = (r, c)$,

$$L_{(r,c)}(s, d) = \begin{cases} (r, s), & s \neq r, \\ (r, r), & s = r,\ d = c, \\ (r, \overline{cd}), & s = r,\ d \neq c. \end{cases}$$

Consequently $L_{(r,c)}(M) = F_r$.

Proof. The three cases are exactly the cross-row projection, diagonal anchoring, and same-row Steiner branches of RP. They produce all three points of F_r. □

Theorem 10.2. For every $x \in M$,

$$\operatorname{rank} L_x = 3, \qquad \operatorname{Im} L_x = F_{\mathrm{row}(x)}.$$

Moreover,

$$\dim \operatorname{span}\{L_x : x \in M\} = 9.$$

Proof. Proposition 10.1 shows that each L_x maps onto the three-point row fiber $F_{\mathrm{row}(x)}$, hence has rank three after linearization. Operators with different target rows have disjoint codomain support. For a fixed row, the three restrictions to the row fiber are linearly independent; in normalized row 0 they contain the matrices

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus each row contributes dimension 3, and the total dimension is 9. The `L3/` operator rank and spectrum audit tables record the nine-row calculation. □

Assume $\operatorname{char} k \neq 2$.

Theorem 10.3. If $x = (r, r)$, then

$$\mu_{L_x}(t) = t(t-1)(t+1)$$

and L_x is semisimple. If $x = (r, c)$ with $r \neq c$, then

$$\mu_{L_x}(t) = t^2(t-1)(t+1),$$

with zero-primary Jordan structure $J_2(0) \oplus J_1(0)^{\oplus 5}$.

Proof. Every $L_{(r,c)}$ maps $k[M]$ onto kF_r, so the nonzero spectral part is the restriction to kF_r. For $c = r$, this restriction fixes (r, r) and swaps the two off-diagonal basis vectors. For $c \neq r$, the restriction has eigenvalues $0, 1, -1$, and its one-dimensional kernel lies inside the image, producing exactly one length-two zero block. The operator-spectrum table audits the calculation for all nine translations. □

10.2 Left-translation semigroup

Let $\mathcal{S}_L = \langle L_x : x \in M\rangle^+$ be the semigroup generated by nonempty words in the nine left translations. The identity map is not included. It is a semigroup under composition, and the unital monoid obtained by adjoining the identity has one additional element. For $r \in S$, let $\mathcal{W}_r$ be the component of maps whose image lies in F_r.

For fixed r, define

$$\tau_{r,c}(d) = \begin{cases} r, & d = c, \\ \overline{cd}, & d \neq c, \end{cases} \qquad \mathcal{P}_r = \langle \tau_{r,c} : c \in S\rangle.$$

In normalized row 0,

$$\mathcal{P}_0 = \{012, 021, 200, 100, 011, 022\}.$$

For $\psi \in \mathcal{P}_r$ and $c \in S$, set

$$N_{r,\psi,c}(s, d) = \begin{cases} (r, \psi(\tau_{r,c}(d))), & s = r, \\ (r, \psi(s)), & s \neq r, \end{cases}$$

and let $K_{r,t}$ be the constant map to (r, t).

Proposition 10.4 (Fixed-row composition normal form)**.** For every $r \in S$,

$$\mathcal{W}_r = \{N_{r,\psi,c} : \psi \in \mathcal{P}_r,\ c \in S\} \cup \{K_{r,t} : t \in S\}.$$

There are 14 nonconstant maps of the first type and 3 constants; hence $|\mathcal{W}_r| = 17$.

Proof. A nonempty composition has a leftmost generator. If all later generators have the same target row r, the word has the displayed $N_{r,\psi,c}$ form. If some later generator has target row different from r, the leftmost row-r generator sees a cross-row input and the composition becomes constant. The normalized finite count $18 - 4 = 14$ for the nonconstant forms is audited in the left-translation semigroup table. □

Set

$$W_r = \operatorname{span}_k \mathcal{W}_r \subseteq \operatorname{End}_k(k[M]).$$

Theorem 10.5. The semigroup generated by the left translations has size

$$|\mathcal{S}_L| = 51.$$

The identity map is not contained in it; adjoining the identity gives the unital generated monoid of size 52. More precisely,

$$\mathcal{S}_L = \mathcal{W}_0 \sqcup \mathcal{W}_1 \sqcup \mathcal{W}_2, \qquad |\mathcal{W}_r| = 17.$$

The linear span $k\mathcal{S}_L$ decomposes as

$$k\mathcal{S}_L = W_0 \oplus W_1 \oplus W_2, \qquad \dim W_r = 9, \qquad \dim k\mathcal{S}_L = 27,$$

and satisfies $W_r W_s \subseteq W_r$.

Proof. Proposition 10.4 gives $|\mathcal{W}_r| = 17$, and the three target-row components are disjoint. Every nonempty word has image contained in a single row fiber, so the identity map on M is not a nonempty word. By definition, $W_r = \operatorname{span}_k \mathcal{W}_r$. The leftmost-variable row law gives $\mathcal{W}_r\mathcal{W}_s \subseteq \mathcal{W}_r$, hence $W_r W_s \subseteq W_r$ after linearization. The linear dimension claim is the row-wise basis-reduction audit recorded in the `L3/` left-translation-semigroup and product-law tables. □

10.3 Row-fiber algebra

Assume $\operatorname{char} k \neq 2$. For each row r, define

$$A_r = \langle L_{(r,c)}|_{kF_r} : c \in S \rangle \subseteq \operatorname{End}(kF_r).$$

Proposition 10.6. The algebra A_r has dimension 5. Its center is kI, its radical is two-dimensional and square-zero, and

$$A_r / \operatorname{Rad}(A_r) \cong k^3.$$

For $r = 0$, one basis is

$$L_{(0,0)}|_{kF_0}, \quad L_{(0,1)}|_{kF_0}, \quad L_{(0,2)}|_{kF_0}, \quad I, \quad (L_{(0,1)}|_{kF_0})^2.$$

Proof. By σ-equivariance it is enough to work in row 0. The five-dimensional multiplication table, center, radical, quotient, and operator-boundary checks are recorded in the auxiliary fiber-algebra certificate files listed in `SUPPORT_INDEX.txt`. ☐

Theorem 10.7. On the six directed edges,

$$\mathbb{R}[M^\times] \cong \mathbf{1} \oplus \operatorname{sign} \oplus 2 \operatorname{std}.$$

For RP, the right-stabilizer adjacency matrix is

$$A_R = (P_C + P_J)^\top$$

and has spectrum

$$\{2, 0, 0, 0, -1, -1\}.$$

Proof. The six directed edges carry the permutation representation of the directed triangle. The right-stabilizer transitions of RP are exactly C and J, so the right-stabilizer adjacency matrix is $(P_C + P_J)^\top$. The transition and spectrum tables audit the finite calculation. ☐

11 Endomorphisms and intertwiners

For a map $\rho : S \to S$, write $\varphi_\rho(r, c) = (\rho(r), \rho(c))$. For $t \in S$, write $e_t(r, c) = (t, t)$.

For a row r, set

$$c \star_r d = \begin{cases} r, & c = d, \\ \overline{cd}, & c \neq d. \end{cases}$$

Lemma 11.1 (Row-fiber homomorphisms)**.** If $\phi : S \to S$ satisfies $\phi(r) = t$ and

$$\phi(c \star_r d) = \phi(c) \star_t \phi(d) \qquad (c, d \in S),$$

then ϕ is either the constant map to t, or a permutation of S sending r to t.

Proof. After translating to $r = t = 0$, write $\phi(1) = u$, $\phi(2) = v$. The relations $0 \star_0 1 = 2$ and $0 \star_0 2 = 1$ force

$$(u, v) = (0, 0), \quad (1, 2), \quad (2, 1),$$

which are respectively the constant solution and the two permutations fixing 0. ☐

Theorem 11.2. The endomorphism monoid of the selected RP magma is

$$\mathrm{End}(M, \cdot) = \{\varphi_\rho : \rho \in \mathrm{Sym}(S)\} \cup \{e_0, e_1, e_2\}.$$

Proof. The idempotents of RP are exactly the diagonal points (r, r). Thus an endomorphism sends row fibers to row fibers over a map $\rho : S \to S$, and its restriction to each row fiber is governed by Lemma 11.1. The cross-row rule forces all nonconstant row restrictions to be the same permutation ρ. If ρ fails to be injective, the cross-row equations force all affected row restrictions to collapse, and then the third row collapses as well; the map is one of the e_t. Conversely, simultaneous permutations and diagonal collapses preserve the three branches of the RP multiplication. □

For an endomorphism φ, let Hom_φ be the space of intertwiners satisfying

$$TL_x = L_{\varphi(x)}T \qquad (x \in M).$$

Let $V = k[M]$, and let

$$\varepsilon : V \to k, \qquad \varepsilon\left(\sum_{m\in M} a_m e_m\right) = \sum_{m\in M} a_m$$

be the total augmentation functional. For $r \in S$, set

$$d_r = e_{(r,r)}, \qquad o_r = \sum_{c\neq r} e_{(r,c)}, \qquad U_r = \mathrm{span}\{d_r, o_r\}.$$

Here $u\varepsilon$ denotes the rank-one operator $v \mapsto \varepsilon(v)u$.

Proposition 11.3 (Constant intertwiners)**.** For each diagonal collapse $e_r(x) = (r, r)$,

$$\mathrm{Hom}_{e_r} = \{u\varepsilon : u \in U_r\} = \left\{ \left(\alpha e_{(r,r)} + \beta \sum_{c\neq r} e_{(r,c)}\right) \varepsilon : \alpha, \beta \in k \right\}.$$

In particular, $\dim \mathrm{Hom}_{e_r} = 2$.

Proof. Each L_x is induced by a map $M \to M$, hence every column of its matrix has sum one and $\varepsilon L_x = \varepsilon$. If $u \in \mathrm{Fix}(L_{(r,r)})$, then

$$(u\varepsilon)L_x = u\varepsilon \qquad \text{and} \qquad L_{(r,r)}(u\varepsilon) = (L_{(r,r)}u)\varepsilon = u\varepsilon,$$

so $u\varepsilon \in \mathrm{Hom}_{e_r}$.

Conversely, if $T \in \mathrm{Hom}_{e_r}$, then

$$TL_x = L_{(r,r)}T = TL_y \qquad (x, y \in M),$$

so T kills

$$\mathcal{D} = \mathrm{span}\{(L_x - L_y)v : x, y \in M,\ v \in V\}.$$

Because $\varepsilon L_x = \varepsilon$, one has $\mathcal{D} \subseteq \ker \varepsilon$. The reverse inclusion is finite: the graph generated by relations $x \cdot z \sim y \cdot z$, with common right argument z, is connected. The `L3/` linear-intertwiner audit records an explicit eight-edge spanning tree, so the differences in $\mathcal{D}$ span the augmentation-zero hyperplane. Thus $\mathcal{D} = \ker \varepsilon$, and T factors uniquely as $T = u\varepsilon$.

Substitution into the intertwining equation gives $L_{(r,r)}u = u$. Finally, $L_{(r,r)}$ has image in the row fiber $F_r = \mathrm{span}\{e_{(r,c)} : c \in S\}$; on this fiber it fixes $e_{(r,r)}$ and swaps the two off-diagonal basis vectors. Therefore

$$\mathrm{Fix}(L_{(r,r)}) = \mathrm{span}\left\{ e_{(r,r)}, \sum_{c \neq r} e_{(r,c)} \right\} = U_r,$$

which proves the formula. □

Proposition 11.4 (Intertwiner classification)**.** Let P_ρ be the permutation operator on V induced by φ_ρ. For $\rho \in \mathrm{Sym}(S)$,

$$\mathrm{Hom}_{\varphi_\rho} = kP_\rho, \qquad \dim \mathrm{Hom}_{\varphi_\rho} = 1.$$

For the diagonal collapse e_r,

$$\mathrm{Hom}_{e_r} = U_r\varepsilon, \qquad \dim \mathrm{Hom}_{e_r} = 2.$$

Thus the total audited intertwiner dimension is

$$6 \cdot 1 + 3 \cdot 2 = 12.$$

Proof. For an automorphism φ_ρ, the operator P_ρ satisfies

$$P_\rho L_x = L_{\varphi_\rho(x)} P_\rho \qquad (x \in M),$$

so $kP_\rho \subseteq \mathrm{Hom}_{\varphi_\rho}$. The `L3/` linear-intertwiner audit records the defining linear constraints and gives $\dim \mathrm{Hom}_{\varphi_\rho} = 1$ for the six automorphism spaces; hence equality holds. The formula for Hom_{e_r} and its dimension are Proposition 11.3. Summing over the six automorphisms and three diagonal collapses gives the displayed total. □

12 Binary term operations

A binary term operation of RP is a function $M^2 \to M$ realized by a term in variables x, y. The binary part of the term clone is denoted $\mathrm{Clo}_2(\mathsf{RP})$.

By the leftmost-variable row lemma, every binary term has a leftmost variable $\ell(f) \in \{x, y\}$. Order the inputs as (m, n), where m is the leftmost variable and n is the other variable. On the same-row sector, normalize the common row to 0 and define

$$s_f(i, j) = \mathrm{col}\big(f(m = (0, i), n = (0, j))\big).$$

On the cross-row sector, normalize the leftmost-variable row to 0 and the other input row to 1; the output column is independent of the other input column, so

$$u_f(i) = \mathrm{col}\big(f(m = (0, i), n = (1, j))\big).$$

The opposite cross-row direction is determined by $u_f^-(i) = -u_f(-i)$.

Proposition 12.1 (Binary sector reduction)**.** Every binary term operation has a normal form

$$\Theta(f) = \big(\ell(f), s_f, u_f\big).$$

The same-row component $s_f : S^2 \to S$ and cross unary component $u_f : S \to S$ determine the term operation on all row sectors.

Proof. The leftmost-variable coordinate is Lemma 9.1. Same-row inputs remain in the common input row. Cross-row inputs are governed by the leftmost-variable row and a unary leftmost-variable column component; the opposite direction is forced by the RP-compatible relabelling exchanging the two non-leftmost directions. □

The realized same-row components form a 21-element set $\mathcal{S}_{21}$. The realized cross unary components form a 15-element set $\mathcal{U}_{15}$. The candidate normal forms are

$$\{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}.$$

Lemma 12.2 (Binary term certificate protocol)**.** The binary term theorem is supported by four finite obligations: T2-SIG records $\mathcal{S}_{21}$ and $\mathcal{U}_{15}$; T2-REAL realizes every triple; T2-TRANS checks closure of the two factor transition systems; T2-MULT reconstructs the full 630×630 composition table from the leftmost variable and factor transitions.

Theorem 12.3. The binary part of the term clone of RP is finite, and the sector map gives a normal-form bijection

$$\Theta : \mathrm{Clo}_2(\mathsf{RP}) \xrightarrow{\sim} \{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}.$$

The composition law is reconstructed by the T2-MULT certificate. Consequently

$$|\,\mathrm{Clo}_2(\mathsf{RP})| = 2 \cdot 21 \cdot 15 = 630.$$

Proof. Proposition 12.1 gives injectivity of Θ. The realization certificate supplies one realizing term for every triple, hence Θ is surjective. The T2-MULT certificate reconstructs the composition law from the finite transition tables. Hence the cardinality is $2 \cdot 21 \cdot 15$. □

This is the binary layer of the term clone. The full term clone is left for the open-problem list.

13 Matrix realization and row-sum contraction

Under the matrix realization

$$k[M] \cong \mathrm{Mat}_3(k), \qquad e_{(r,c)} \leftrightarrow E_{rc},$$

the finite RP multiplication and ordinary matrix multiplication become two bilinear structures on the same nine-dimensional vector space. The row-sum contraction records the cross-row branch of RP on matrix units.

Proposition 13.1. Under $k[M] \cong \mathrm{Mat}_3(k)$, the row-sum contraction extension of the cross-row RP rule is

$$\mu_{cross}(A, B) = AJ_3B^\top = (A\mathbf{1})(B\mathbf{1})^\top.$$

On matrix units with cross rows,

$$E_{ij}J_3E_{kl}^\top = E_{ik}.$$

Proof. Since $J_3 = \mathbf{1}\mathbf{1}^\top$,

$$E_{ij}J_3E_{kl}^\top = (e_ie_j^\top)(\mathbf{1}\mathbf{1}^\top)(e_le_k^\top) = E_{ik}.$$

For $i \neq k$, this is exactly the cross-row RP rule $(i, j) \cdot (k, l) = (i, k)$. The `L3/` matrix-realization audit tables record the basis-level realization. □

Thus the matrix realization expresses the same column-independence selected by $\mathcal{H}_{acc} = 0$: cross-row multiplication factors through row sums. The same-row branch remains the Steiner/diagonal row-fiber rule.

14 Operator interpretation of the H-statistic

The local Hamming-defect identity of Section 4 has an operator interpretation in the left-regular semigroup. For a normalized local index q, D_C compares the two stepwise continuation signatures, while D_L compares the two collapsed left translations.

Theorem 14.1 (Hamming-defect operator identity)**.** Let $\omega \in \Omega'$ be a completed normal-form magma and let $q = (b, t, a, e, f) \in S^5$. Define

$$xy, \qquad yz, \qquad z_R, \qquad ep_L, \qquad ep_R$$

from ω as in Theorem 4.1. Then

$$\frac{h_q(\omega)}{2} = D_C(t, xy, f; b, a, z_R) - D_L(ep_L, ep_R).$$

For brevity write

$$D_C(q; \omega) = D_C(t, xy, f; b, a, z_R), \qquad D_L(q; \omega) = D_L(ep_L, ep_R).$$

After selection, take $\omega = \mathsf{RP}$ and write the normalized triple as

$$X = (0, a), \qquad Y = (b, e), \qquad Z = (t, f).$$

Then

$$XY = (0, xy), \qquad YZ = (b, z_R),$$

and

$$(XY)Z = (0, ep_L), \qquad X(YZ) = (0, ep_R).$$

With the convention $AB = A \circ B$, the continuation distance $D_C(q; \mathsf{RP})$ is the Hamming-distance contrast between the normalized signatures of

$$L_{XY} L_Z \quad \text{and} \quad L_X L_{YZ},$$

whereas $D_L(q; \mathsf{RP})$ is the Hamming-distance contrast between

$$L_{(XY)Z} \quad \text{and} \quad L_{X(YZ)}.$$

As maps, all four operators lie in $\mathcal{W}_{\mathrm{row}(X)}$; after linearization they lie in $W_{\mathrm{row}(X)}$.

Define

$$I_{tot}(\omega) = 6 \sum_{q \in S^5} D_C(q; \omega), \qquad B_{tot}(\omega) = 6 \sum_{q \in S^5} D_L(q; \omega).$$

Thus $H_{tot}(\omega) = I_{tot}(\omega) - B_{tot}(\omega)$. For RP,

$$I_{tot} = 8904, \qquad B_{tot} = 1884, \qquad H_{tot} = 7020, \qquad N_- = 0.$$

The finalist RC has the same nonnegative local Hamming-defect profile.

Proof. The scalar identity is exactly Theorem 4.1 divided by 2: D_C is the sum of the nine continuation disagreement indicators, D_L is the sum of the nine collapsed left-translation disagreement indicators, and h_q is twice their signed difference.

For $\omega = \mathsf{RP}$, the displayed normalized triple identifies the two continuation signatures with the signatures of $L_{XY} L_Z$ and $L_X L_{YZ}$, under the convention $AB = A \circ B$. Likewise the collapsed signatures are the left translations by the endpoints $(XY)Z$ and $X(YZ)$. The row-target assertion follows from the left-translation-semigroup law $W_r W_s \subseteq W_r$. The `L3/` Hamming-defect realization audit records the RP/RC comparison and verifies the displayed values. The global range status is Theorem 4.2. ☐

15 Associated compact Lie groups

The matrix realization

$$k[M] \cong \mathrm{Mat}_3(k), \qquad e_{(r,c)} \leftrightarrow E_{rc},$$

places the selected finite magma on the same nine-dimensional vector space as the ordinary matrix algebra. An *associated compact Lie group* is a compact Lie group attached to a specified linear structure on this shared carrier. The ordinary matrix and Lie structures used here are standard [18, 12, 17].

Over fields of characteristic different from 3, ordinary matrix multiplication gives

$$\mathfrak{gl}(3,k) = kI \oplus \mathfrak{sl}(3,k), \qquad \mathrm{Mat}_3(k) = kI \oplus \mathfrak{sl}(3,k)$$

as vector spaces. After complex compact form, this is the usual central $\mathfrak{u}(1)$ plus $\mathfrak{su}(3)$ picture.

Over $\mathbb{R}$, the compact three-dimensional branch

$$\mathfrak{so}(3) = \{X \in \mathfrak{sl}(3,\mathbb{R}) : X^\top = -X\}$$

is stable under simultaneous permutation conjugation. A split comparator is $\mathfrak{sl}(V_{std}) \subset \mathfrak{sl}(3,\mathbb{R})$, where

$$V_{std} = \{v : v_0 + v_1 + v_2 = 0\}.$$

Compactness selects the $\mathfrak{so}(3)$ branch among these three-dimensional $\mathrm{Sym}(S)$-stable carrier branches.

Theorem 15.1. In the real/complex compact-form setting, the matrix realization, together with the RP-compatible $\mathrm{Sym}(S)$-action, exhibits the following associated compact types:

$$SU(3), \qquad SU(2), \qquad U(1).$$

Here $SU(3)$ is attached to the ordinary $\mathfrak{sl}(3)$ matrix package, $SU(2)$ is the simply connected compact group of the selected $\mathfrak{so}(3)$ branch, and $U(1)$ is attached to the central scalar line.

Proof. The matrix-unit carrier gives Mat_3, hence the ordinary $\mathfrak{sl}(3)$ matrix package and its compact form $SU(3)$. The $\mathrm{Sym}(S)$-stable compact branch $\mathfrak{so}(3)$ gives $SO(3)$, whose simply connected compact cover is $SU(2)$. The central scalar line gives the compact circle $U(1)$. The `L3/` compact-group realization audit records these associated compact types. $\square$

16 Signed symplectic lift

This realization uses the finite C/J right-stabilizer geometry selected in Part II. The continuous statement is a signed projective symplectic realization of that six-edge geometry and completes the matrix realization part of the structural profile [22].

Let $V = \mathbb{R}^6$ have ordered basis

$$(e_{01}, e_{12}, e_{20}, e_{10}, e_{02}, e_{21})$$

and symplectic matrix

$$\Omega = \begin{pmatrix} 0 & I_3 \\ -I_3 & 0 \end{pmatrix}.$$

Set

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$\widehat{C} = \begin{pmatrix} P & 0 \\ 0 & P \end{pmatrix}, \qquad \widehat{J}_0 = \begin{pmatrix} 0 & R \\ R & 0 \end{pmatrix}, \qquad \widehat{J}_s = \begin{pmatrix} 0 & R \\ -R & 0 \end{pmatrix}.$$

The unsigned $\widehat{J}_0$ induces edge reversal but is anti-symplectic; the signed $\widehat{J}_s$ induces the same projective edge reversal and is symplectic.

Theorem 16.1. The finite C/J right-stabilizer directed-edge geometry admits the signed projective dihedral lift displayed above. The unsigned reversal is anti-symplectic, while the signed reversal is symplectic. The finite word

$$C, J, C, C, J$$

visits all six directed edges and has a signed symplectic realization.

Proof. Direct multiplication gives

$$\widehat{C}^T \Omega \widehat{C} = \Omega, \qquad \widehat{J}_0^T \Omega \widehat{J}_0 = -\Omega, \qquad \widehat{J}_s^T \Omega \widehat{J}_s = \Omega,$$

and

$$\widehat{C}^3 = I, \quad \widehat{J}_s^2 = -I, \quad \widehat{J}_s \widehat{C} \widehat{J}_s^{-1} = \widehat{C}^{-1}.$$

Hence the projective classes satisfy the dihedral relations in $\mathrm{PSp}(V, \omega)$. On coordinate lines, $\widehat{C}$ acts as (01 12 20)(10 02 21), and $\widehat{J}_s$ acts as edge reversal. Starting from 01, the word gives

$$01 \mapsto 12 \mapsto 21 \mapsto 10 \mapsto 02 \mapsto 20.$$

The signed-lift matrices and word audit are recorded in the `L3/` signed-symplectic realization tables. ☐

The signed lift completes the post-selection structural profile of RP.

17 Conclusion

The normal-form parameter space

$$\Omega' \cong S^{21}$$

is mapped by the associativity spectrum and distribution and the path-resolved Hamming defect. The four internal selection conditions

$$\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad (r, r)^2 = (r, r), \qquad \mathrm{StabMaps}_R = \{C, J\}$$

select the right-row projection magma RP. After this selection, Part III records the finite post-selection structural profile: generated semigroup, term clone, endomorphism/intertwiner classification, row-sum contraction matrix realization, Hamming-defect operator identity, associated compact Lie groups, and signed symplectic lift.

The manuscript therefore has the publication chain

$$\Omega' \cong S^{21} \xrightarrow{\text{selection}} \mathsf{RP},$$

$$\mathsf{RP} \xrightarrow{\text{linearization}} \begin{array}{c}\text{finite operator, term,} \\ \text{and carrier structural profile}\end{array}.$$

Companion material

A Finite certificate protocol

The finite support package is available in the public GitHub repository:

`https://github.com/burundai-t/finite-row-graded-magmas-over-f3`.

The finite algebraic language follows the standard binary-system, universal-algebra, design, quasigroup, and clone-theory sources [5, 6, 7, 8, 11, 9, 23, 24, 25, 27, 20]. The certificate style is the standard finite-proof pattern: a human-readable reduction is paired with reproducible finite artifacts [1, 2, 13, 15, 16, 21].

component	role	manuscript interface
`L1/`	finite parameter space and associativity reference checks	normal form, associativity spectrum and distribution, endpoint loci, partition function
`L2/`	compact selection and C/J right-stabilizer condition survivor checks	selection conditions, directed-edge right stabilizers, uniqueness theorem
`L3/`	selected-RP operator and realization checks	generated semigroup, Clo_2, matrix realization, associated compact Lie groups, signed symplectic lift
`mathcal_H/`	global path-resolved Hamming defect bundle	local distance reduction, H1–H4 Hamming-defect obligations
repository interface	`README.md`; `SUPPORT_INDEX.txt`	reviewer commands and file-by-file index

The compact table families are:

family	location	purpose
L1 parameter-space distribution	`L1/tables/`	associativity distribution, endpoints, orbit representatives, controlled $\mathcal{H}$ comparison
L2 selection	`L2/tables/`	zero conditional mutual information, zero diagonal entropy, C/J right-stabilizer condition survivor separation
L3 linearization profile	`L3/tables/`	left-translation semigroup, binary part of the term clone, matrix realization, signed symplectic lift, fiber algebra
path-resolved Hamming defect	`mathcal_H/`	H1 unrestricted range, H2 nonnegative-local-locus UNSAT, H3 witnesses, H4 maximal-layer classifier

A.1 Global path-resolved Hamming defect certificate handles

The global range theorem for the path-resolved Hamming defect is checked by a separate `mathcal_H/` bundle. At manuscript level the implication is

$$\text{local distance reduction} + \text{H1–H4 certificate obligations} \Longrightarrow \text{global range theorem.}$$

handle	bundle location	theorem component
HF-RED, HF-CERT-IF	bundle theorem interface	local distance reduction and local signature formulas
H1	`h1/`	unrestricted full-parameter-space range $H_{\text{norm}} \in [-378, 1217]$
H2	`h2/`	nonnegative-local-locus exclusion at $H_{\text{norm}} \geq 1171$
H3	embedded witness verifier	RP and RC at $H_{\text{norm}} = 1170$, $N_{-} = 0$
H4	`h4/`	twelve-point maximal layer with negative local contributions at $H_{\text{norm}} = 1217$

The H2 obligation is the finite UNSAT statement

$$(h_q \geq 0 \ \forall q \in S^5) \wedge H_{\text{norm}} \geq 1171$$

on the depth-9 live search locus:

$$18540/18540 \text{ UNSAT}, \qquad \text{SAT} = 0, \qquad \text{UNKNOWN} = 0.$$

The light static coverage verifier for the accepted H2 source artifacts is:

```
cd mathcal_H/h2
PYTHONDONTWRITEBYTECODE=1 python3 verify_h2_coverage_light.py
```

A.2 Dependency summary

result family	support role
normal form and selection proof	hand finite arguments with L2 anchor checks
associativity spectrum and distribution	block reduction plus L1 endpoint and partition-function tables
global path-resolved Hamming defect	local distance reduction plus H1–H4 certificate bundle
generated semigroup and row-fiber algebra	analytic leftmost-variable row reductions plus L3 multiplication tables
endomorphisms and intertwiners	finite row-fiber classification with linear intertwiner audit
binary part of the term clone	sector reduction plus realization and transition certificates
matrix, compact Lie group, and symplectic realizations	matrix-realization calculations supported by L3 realization audits

B One-page structural profile of RP

The selected point is

$$\mathsf{RP} = (a, b, d) = (1, 2, 000).$$

It has the finite economy and compression profile used by the selection theorem:

$$\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad (r, r)^2 = (r, r), \qquad a \neq b, \qquad d = 000.$$

Thus its cross-row rule is column-independent, its same-point diagonal is compressed and anchored, and its two cross-row anchors are nondegenerate. The Landauer/Shannon provenance of the first two conditions is zero conditional mutual information and zero diagonal entropy [19, 3, 4, 26, 10].

Proposition B.1 (Nondegenerate anchors)**.** Among the nine column-independent rules $h_{a,b}$, the condition $a \neq b$ leaves exactly six rules.

Proof. There are nine pairs $(a, b) \in S^2$, and the three pairs (a, a) are excluded. ☐

Proposition B.2 (Local associativity finalists)**.** With $d = 000$, the six nondegenerate column-independent rules have

$$\mathrm{Assoc}_{000} = 219$$

exactly at $(a, b) = (1, 2)$ and $(2, 1)$, namely RP and RC. The other four nondegenerate rules have $\mathrm{Assoc}_{000} = 273$.

Proof. By Proposition 3.3, on the column-independent, $d = 000$ stratum,

$$\mathrm{Assoc}_{000}(h_{a,b}) = 3\,(73 + 18[a = 0] + 18[b = 0] + 54[a = b])\,.$$

On $a \neq b$, this is minimized exactly when neither anchor is 0, giving $(1, 2)$ and $(2, 1)$. ☐

The selected directed-edge feature is the head-row right-stabilizer condition.

Lemma B.3 (Right-stabilizer profile)**.** For RP,

$$\mathrm{StabMaps}_{R,\mathsf{RP}} = \{C, J\}.$$

The row-complementary point $\mathsf{RC} = (a, b, d) = (2, 1, 000)$ has

$$\mathrm{StabMaps}_{R,\mathsf{RC}} = \{C^{-1}, C^{-1}J\}.$$

Proof. Let $x = (r, c) \in M^{\times}$. For RP, the cross-row rule is $h(r_x, r_y) = r_y$, so right-stabilization is $r_y = c$. The two right-stabilizer neighbours are (c, r) and $(c, \overline{rc})$, namely $J(x)$ and $C(x)$.

For RC, the cross-row rule is $h(r_x, r_y) = \overline{r_x r_y}$. The right-stabilizer equation $\overline{r r_y} = c$ gives $r_y = \overline{rc}$. The two right-stabilizer neighbours are $(\overline{rc}, r)$ and $(\overline{rc}, c)$, namely $C^{-1}(x)$ and $C^{-1}J(x)$. ☐

This one-page structural profile records RP at the selection boundary: zero conditional mutual information, zero diagonal entropy, anchored self-action, nondegenerate anchors, local associativity finalist value, and the head-row right-stabilizer pair $\{C, J\}$. The adjacent row-complement finalist has the drifted right-stabilizer pair $\{C^{-1}, C^{-1}J\}$.

C Information-theoretic and structural interpretation

The four selection conditions (together with the supporting structural properties of the selected magma RP) admit the following information-theoretic and structural reading:

Mathematical condition	Information-theoretic / structural interpretation
$\mathcal{H}_{\rm acc} = 0$	Do not read hidden columns; avoid Landauer-facing erasure of hidden information.
$(a \neq b)$, for RP $((a, b) = (1, 2))$	Non-triviality: the output depends on the right row direction and does not collapse to a constant response.
$H_{\rm diag} = 0$	Minimize Shannon entropy on the diagonal — do not distinguish diagonal types where distinction is unnecessary.
$\min \mathrm{Assoc}_{000}$ / low local associativity	The order of application has meaning (the sequence of operations carries information).
C/J right-stabilizer condition and signed symplectic lift	Directed-edge structure is preserved without drift; at the linear level, phase-volume preservation appears.
$((r, c)^2 = (r, r))$	Self-application returns to the element's own row-fixed point.

These six readings can be distilled into two complementary axes of a single principle:

$$\boxed{\text{Economize, but do not descend into triviality.}}$$

$$\boxed{\text{Preserve structure and information — without unnecessary expenditure.}}$$

D Open problems and extensions

The following problems are subsequent work beyond the theorem package proved in this manuscript.

D.1 A functorial relation between RP and $\mathfrak{gl}(3)$

The shared vector space

$$k[M] \cong \mathrm{Mat}_3(k)$$

supports the finite RP multiplication and ordinary matrix multiplication. The row-sum contraction

$$\mu_{cross}(A, B) = AJ_3B^{\top}$$

aligns the cross-row branch with matrix rank-one contraction. The next problem is to identify a universal or functorial construction from the finite magma to the matrix/Lie package. The matrix and Lie background used here is standard material [18, 12, 17].

D.2 Symplectic realization of the C/J geometry

The C/J right-stabilizer geometry has an explicit signed symplectic lift. The next layer is the associated continuous symplectic geometry: canonical (q, p)-charts, Hamiltonian paths, and coordinate-free separation between head-row reversal and row-complement drift. This is the natural continuation of the finite lift in the language of symplectic geometry [14, 22].

D.3 Unified selection principle

The uniqueness theorem uses four finite conditions: zero conditional mutual information, diagonal entropy zero, idempotence on the diagonal orbit, and C/J right-stabilizer condition. A sharper formulation would derive these conditions from one natural functional whose global minimizer on Ω' is the RP representative.

D.4 Full term clone

The binary part of the term clone has the normal-form bijection

$$\Theta : \mathrm{Clo}_2(\mathsf{RP}) \xrightarrow{\sim} \{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}, \qquad |\,\mathrm{Clo}_2(\mathsf{RP})| = 630.$$

The full term clone remains to be classified. The relevant background is the classical clone-theoretic framework [24, 25, 27, 20].

D.5 Larger carriers

Natural extensions include analogues over larger finite fields or cyclic sets and directed-edge C/J geometries beyond the triangle. The standard combinatorial-design and quasigroup references provide the ambient language for these variants [11, 9, 23].

D.6 Relaxing AX2

The present parameter space fixes the same-row Steiner rule. The relaxed parameter space asks whether that same-row rule can itself be selected intrinsically.

Note on AI assistance

This work was developed with assistance from large language models. The author specified the conceptual framework, constraints, and target properties of the objects under study. Within this framework, AI tools were used for computational exploration, code drafting, and structuring intermediate results. All mathematical definitions, statements, and theorems presented in the paper were established and independently verified by the author. The finite claims are backed by exhaustive checks recorded in the accompanying support package.

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